ITER Time-Dependent Plasma Simulation

This notebook specifically demonstrates a full-discharge time-dependent simulation spanning t = 50 s to t = 350 s, beginning at a converged self-consistent plasma state and evolving current diffusion, equilibrium, transport, pedestal, and sawtooth physics simultaneously, with a plasma current (Ip) feedback controller active throughout.

Simulation Time-Dependent

The key distinction between a stationary and time-dependent FUSE simulation is how ini parameters are specified. In a stationary run, ini fields are scalar numbers. In a time-dependent run, they are Julia anonymous functions of the form:

```
ini.nb_unit[1].power_launched = t -> (1 .+ ramp((t - 100) / 100)) * 16.7e6
ini.ec_launcher[1].power_launched = t -> (1 .+ ramp((t - 100) / 100)) * 10E6

ini.equilibrium.ip = t -> sequence(t, [(0.0, 0.0), (10.0, 13.0E6), (100.0, 13.0E6), (200.0,
15.0E6)]);
```

When FUSE.init() is called with these time-dependent ini values, it evaluates each function across the pulse_shedule_time_basis time grid (0 to 300s, step 1s) and stores the resulting time series in dd.pulse_schedule. The ActorDynamicPlasma then reads dd.pulse_schedule at each time step to know what Ip to target, how much heating power to apply, etc.

```
ini, _ = FUSE.case_parameters(:ITER; init_from=:scalars, time_dependent=true);

ini.time.pulse_shedule_time_basis = range(0, 300; step=1.0)
ini.time.simulation_start = 50.0;

rampup_ends = 10.0
```

2.3 Pulse Shaping Functions

FUSE provides a library of normalized pulse-shaping functions. Each function maps a normalized time argument $\tau \in [0,1]$ to a normalized output $y \in [0,1]$. The physical time is converted to normalized time by dividing by a characteristic time scale.

Function	Mathematical Form	Description	Typical Use
ramp(τ)	clamp(τ, 0, 1)	Linear ramp: 0→1 as τ goes 0→1. Most common.	ip ramp-up
step(τ)	H(τ - 0.5) where H = Heaviside	Step function: 0 for τ<0.5, 1 for τ≥0.5	Sudden transitions
pulse(τ)	ramp(2τ) - ramp(2τ-1)	Trapezoidal pulse: rises, holds, falls	Transient heating bursts
trap(τ,f)	Trapezoidal with ramp fraction f	f controls rise/fall fraction	Custom flat-tops
gaus(τ,n)	exp(-((τ-0.5)/σ)^2n)	Generalized Gaussian of order n	Smooth peaked profiles
beta(τ,mode)	Beta distribution shape	Asymmetric bell curve	Profile shaping

Function	Mathematical Form	Description	Typical Use
sequence(t, pts)	Piecewise linear interpolation	List of (t,y) anchor points	Arbitrary waveforms

2.4 The ip(t) Waveform

The plasma current waveform defined in this notebook goes through three distinct phases:

```
ini.equilibrium.ip = t -> sequence(t, [(0.0, 0.0), (10.0, 13.0E6), (100.0, 13.0E6), (200.0,
15.0E6)]);
```

```
ini.equilibrium.ip = t -> sequence(t, [
   (0.0,   0.0   ),   # plasma breakdown: zero current
   (10.0,  13.0E6),   # end of ramp-up: 13 MA reached
   (100.0, 13.0E6),   # flat-top at 13 MA for 90 s
   (200.0, 15.0E6),   # second ramp to 15 MA (Q-enhancement)
])                    # held at 15 MA for remaining duration
```

- Phase 1 (t = 0–10 s): Ramp-up phase. Ip rises linearly from 0 to 13 MA in 10 s. This is the current ramp phase where the plasma transitions from a limiter to a diverted configuration.
- Phase 2 (t = 10–100 s): First flat-top at 13 MA. The plasma is in steady-state operation with the initial heating powers applied.
- Phase 3 (t = 100–200 s): Second ramp from 13 MA to 15 MA over 100 s. This is simultaneous with a doubling of all auxiliary heating powers (NBI, EC, IC, LH).
- Phase 4 (t = 200–300 s): Final flat-top at 15 MA with full heating power (~100 MW auxiliary + ~40 MW alpha). This represents the high-Q ITER operating point.

The simulation begins at t = 50 s (ini.time.simulation_start = 50.0), which is in the middle of the first flat-top phase. The pulse schedule is defined out to t = 300 s but the dynamic run extends to t = 350 s.

2.5 Heating Power Waveforms

All auxiliary heating systems follow the same temporal pattern — a baseline level from t = 0 that doubles as the Ip ramps to 15 MA:

```
# Pattern: starts at P₀, increases to 2P₀ as Ip ramps to 15 MA
ini.nb_unit[1].power_launched = t -> (1 .+ ramp((t - 100) / 100)) * 16.7e6
ini.ec_launcher[1].power_launched = t -> (1 .+ ramp((t - 100) / 100)) * 10E6
ini.ic_antenna[1].power_launched = t -> (1 .+ ramp((t - 100) / 100)) * 12E6
ini.lh_antenna[1].power_launched = t -> (1 .+ ramp((t - 100) / 100)) * 5E6
```

Parameter	Value / Description
NBI (Neutral Beam)	16.7 MW → 33.4 MW at 1 MeV beam energy
EC (Electron Cyclotron)	10 MW → 20 MW, deposited at ρ ≈ 0.37
IC (Ion Cyclotron)	12 MW → 24 MW, minority heating
LH (Lower Hybrid)	5 MW → 10 MW, primarily current drive
Total auxiliary at t=50s	~60 MW
Total auxiliary at t=350s	~104 MW + 40 MW alpha = ~144 MW total

2.6 The Ramp-up Physics Sub-Parameters

Three parameters specifically control the plasma current ramp-up geometry:

```
rampup_ends = 10.0                    # ramp-up ends at t=10 s
ini.rampup.side = :lfs                # ramp-up on Low Field Side (outer wall)
ini.rampup.ends_at = rampup_ends      # 10 s to reach diverted equilibrium
```

```
ini.rampup.diverted_at = rampup_ends * 0.8 # diverted config reached at t=8 s
```

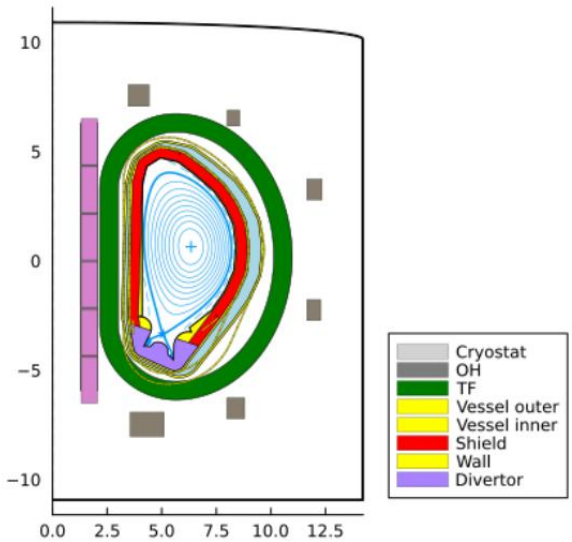
The :lfs setting means the plasma first touches the outer wall (low field side) as a limiter before transitioning to a diverted X-point configuration at t = 8 s. This is the standard ITER ramp-up strategy - the plasma moves from a circular limiter shape to the elongated D-shape diverted configuration.

3. Simulation Workflow

3.1 Hardware Initialization (Cells 1–4)

Parameter	Value / Description
Cell 1	Load Julia packages: Plots (visualization), FUSE (simulation engine)
Cell 2	FUSE.case_parameters(:ITER; init_from=:ods) -> load ITER template from experimental ODS file. Returns ini and act pre-populated for ITER.
Cell 3	dd = IMAS.dd() creates empty data dictionary. FUSE.init(dd, ini, act) populates dd by running: CXbuild (geometry), Sources (all heating initializers).
Cell 4	plot(dd.build) -> visualizes the machine cross-section: Cryostat, OH coil, TF coils, Vacuum vessel (inner/outer), Shield, First wall, Divertor.

The CXbuild actor constructs the complete 2D cross-section geometry from the ODS data, including all structural layers. The result is the familiar ITER D-shaped tokamak cross-section visible in the Cell 4 plot, showing the nested layered structure from the plasma-facing divertor to the outer cryostat.



3.2 Scenario Definition (Cells 5-8)

After checking out the hardware-initialized state, a new ini is created with time_dependent=true and init_from=:scalars. The time-dependent waveforms are then assigned, followed by plot(ini) to verify the pulse schedule visually before committing to the expensive initialization.

FUSE.init(dd, ini, act; initialize_hardware=false) is then called. The initialize_hardware=false flag skips rebuilding the machine geometry (already present from the ODS) and only evaluates the time-dependent ini functions across the pulse_shedule_time_basis grid to populate dd.pulse_schedule.

The Cell 8 STDERR output shows the additional actors that run during this second init: Equilibrium (FRESCO solver), Current (QED current diffusion), and PassiveStructures - these populate the initial plasma state at t = 0.

3.3 Self-Consistent Stationary Solve (Cell 11)

Before starting the time-dependent evolution, the simulation must begin in a physically self-consistent state. ActorStationaryPlasma achieves this through iterative coupling of:

- ActorEquilibrium (FRESCO/TEQUILA): Solves the Grad-Shafranov equation for the 2D equilibrium given pressure and current profiles
- ActorSources (Simple models): Computes EC, IC, LH, NBI, pellet, neutral fueling source profiles
- ActorFluxMatcher: Adjusts density and temperature profiles until the transport model predicts fluxes that match the total power/particle sources - this is the key convergence criterion
- ActorSawteethSource: Models the flattening effect of sawtooth MHD instabilities on the current and temperature profiles

The convergence criterion is that the normalized change in profiles between successive outer iterations falls below 2% (convergence_error = 2E-2). As shown in the output, convergence is achieved in 5 iterations.

```
STDERR

[32mProgress: 100%|██████████████████████████| Time: 0:00:57 ( 0.85  s/it)[39m
[34m            iteration (min 2): 5/10[39m
[34m   required convergence error: 0.02[39m
[34m          convergence history: [0.011259887201354065, 0.024739593589087283,
0.06047188239612959, 0.14628358801130426, 0.23941168198790022][39m
[34m                        stage: N/A[39m
[34m                     Ip [MA]: 13.01464846067517[39m
[34m                   Ti0 [keV]: 13.797276500713583[39m
[34m                   Te0 [keV]: 14.152835709512974[39m
[34m              ne0 [10²⁰ m⁻³]: 0.979324435696635[39m
[34m                  max(zeff): 2.006758785182514[39m
[34m                 ω0 [krad/s]: 10.0[39m
```

3.4 Time-Dependent Evolution (Cell 13–14)

```
act.ActorDynamicPlasma.Nt = 60       # 60 time steps
act.ActorDynamicPlasma.Δt = 300.0    # total time span = 300 s
# → each step covers Δt/Nt = 5 s of physical time
```

At each of the 60 time steps (every 5 s), ActorDynamicPlasma performs:

- 1. Reads the prescribed Ip(t), heating powers, pellet frequency from dd.pulse_schedule at the current time
- 2. Updates equilibrium (Grad-Shafranov solve with evolved current profile)
- 3. Evolves current diffusion (QED equation): $\partial\psi/\partial t = \eta(T) \cdot (\partial^2\psi/\partial\rho^2 + ...) + \eta \cdot j_aux$
- 4. Evolves transport: solves energy and particle balance ODEs for Te(ρ,t), Ti(ρ,t), ne(ρ,t)
- 5. Updates sources (NBI, EC, IC, LH, alpha particle heating from fusion)
- 6. Updates pedestal model (ELM-averaged pedestal Te, ne)
- 7. Checks for sawtooth crash condition and applies profile flattening
- 8. Applies Ip controller: adjusts Vloop to drive Ip toward the reference trajectory

The ip_controller (PID controller in ActorControllerIp) applies loop voltage corrections via the OH solenoid to maintain Ip on the prescribed reference trajectory. This is analogous to real plasma control systems.

```
STDERR

[32mProgress: 100%|██████████████████████████| Time: 0:08:11 ( 0.74  s/it)[39m
[34m       start time: 50.0[39m
[34m         end time: 350.0[39m
[34m             time: 350.0[39m
[34m            stage: PFactive (2/2)[39m
[34m          Ip [MA]: 15.001719020386778[39m
[34m        Ti0 [keV]: 16.51789685627013[39m
[34m        Te0 [keV]: 16.471579625034888[39m
[34m   ne0 [10²⁰ m⁻³]: 1.0199162625131863[39m
[34m        max(zeff): 2.0100507629820425[39m
[34m      ω0 [krad/s]: 10.0[39m
```

4.1 Current Ramp-Up Physics

During the ramp-up phase (t = 0-10 s), the plasma current is driven almost entirely by the OH solenoid (ohmic current). The skin effect in the resistive plasma means that current initially flows only on the plasma surface (outer flux surfaces) and diffuses inward on the resistive diffusion time scale:

$$\tau_resistive = \mu_0 \cdot a^2 / \eta(Te) \approx \mu_0 \cdot a^2 \cdot (Te[eV])^{(3/2)} / (5.2\times10^{-5} \cdot Zeff \cdot \ln \Lambda)$$

```
   Ip [MA]: 13.014648460675 17[39m
 Ti0 [keV]: 13.797276500713583[39m
 Te0 [keV]: 14.152835709512974[39m
```

For ITER with $Te_0 \approx 14$ keV at t = 50 s, this gives τ_resistive ≈ hundreds of seconds - meaning the current profile at the start of the simulation is still relaxing from the ramp-up and has not yet reached its ohmic equilibrium profile. The QED (Quasi-linear Equilibrium Diffusion) actor in FUSE propagates this current diffusion equation self-consistently.

4.2 Bootstrap Current

The bootstrap current is a self-generated neoclassical current that arises from pressure gradients and trapped particle effects in a torus. Its magnitude scales approximately as:

$$j_BS \propto \varepsilon^{(1/2)} \cdot (\partial p/\partial r) / B_pol$$

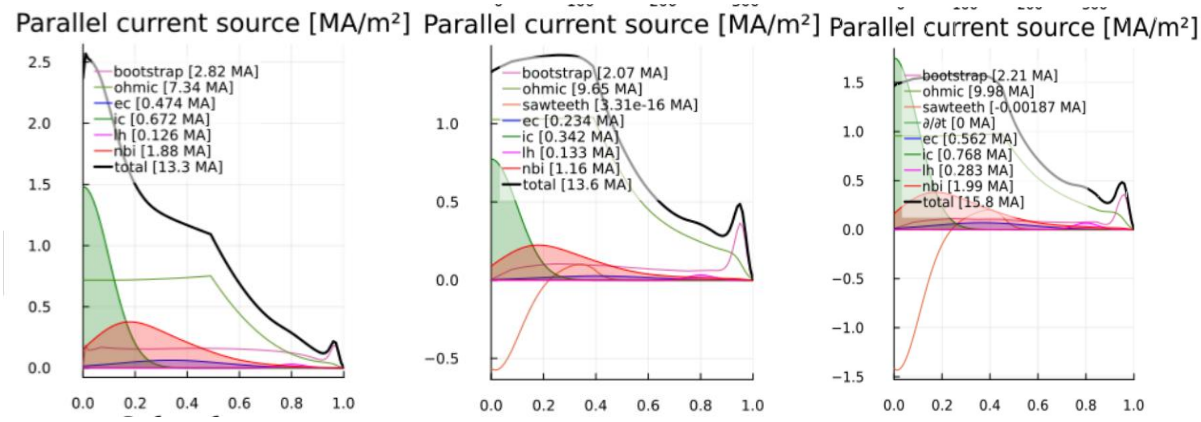


Image	Bootstrap	Ohmic	Total Ip	Stage
Image 1	2.82 MA	7.34 MA	13.3 MA	Cell 10 — **before** stationary solve (raw init)
Image 2	2.07 MA	9.65 MA	13.6 MA	Cell 12 — **after** stationary solve (t = 50 s)
Image 3	2.21 MA	9.98 MA	15.8 MA	Cell 14/15 — **after dynamic run** (t = 350 s)

· **Before stationary solve (Image 1):** 2.82 MA / 13.3 MA = **21.2%**. This is the raw initialization state before the flux-matcher has enforced transport consistency. The initial pressure profile from ini is prescribed and steeper than what the transport model sustains, producing an artificially high bootstrap current. The ohmic current (7.34 MA) is also anomalously low at this stage.

· **After stationary solve at t = 50 s (Image 2):** 2.07 MA / 13.6 MA = **15.2%**. Once ActorStationaryPlasma converges the profiles self-consistently, the transport model flattens the pressure profile slightly (reducing the pressure gradient $\partial p/\partial r$), lowering the bootstrap to 2.07 MA. Simultaneously the ohmic current rises to 9.65 MA to compensate and hold Ip on target. This is the physically credible starting point.

· **After dynamic run at t = 350 s (Image 3):** 2.21 MA / 15.8 MA = **14.0%**. As heating power doubles and Ip ramps to 15 MA, the pressure profile steepens modestly, slightly recovering the bootstrap. But Ip grew faster than the bootstrap, so the bootstrap *fraction* actually decreases from 15.2% to 14.0%. The ohmic current rises to 9.98 MA to carry the additional current load.

4.3 Transport and the H-factor

Cell 19 output:

```
[34m[1mτe[22m[39m[0m → [0m1.8[0m [s]
[34m[1mne_ped[22m[39m[0m → [0m5.9e+19[0m [m⁻³]
[34m[1m<P>[22m[39m[0m → [0m0.2[0m [MPa]
[34m[1mτe_exp[22m[39m[0m → [0m1.47[0m [s]
[34m[1mne_line[22m[39m[0m → [0m8.81e+19[0m [m⁻³]
[34m[1mP0/<P>[22m[39m[0m → [0m3.01
[34m[1mH98y2[22m[39m[0m → [0m0.68
[34m[1m<ne>[22m[39m[0m → [0m7.91e+19[0m [m⁻³]
[34m[1mβn[22m[39m[0m → [0m1.27
[34m[1mH98y2_exp[22m[39m[0m → [0m0.64
```

Plasma energy transport in a tokamak operates through two fundamentally different mechanisms. The first is **neoclassical transport**, a collisional process arising from particle orbit geometry in a

torus, computed in this simulation by the Hirshman-Sigmar model. The second is **anomalous transport**, which is 10 to 100 times larger than neoclassical predictions and is driven by plasma micro-turbulence - specifically Ion Temperature Gradient (ITG) modes and Trapped Electron Modes (TEMs). This anomalous channel is computed by **TGLFNN**, a neural-network surrogate of the quasi-linear TGLF transport model, trained specifically on ITER plasma conditions. These two models serve different roles: Hirshman-Sigmar provides the collisional baseline, while TGLFNN predicts the dominant turbulent heat flux that the flux-matcher must balance against the heating sources at each radial location.

The quality of energy confinement is quantified by the energy confinement time $\tau E = Wp / Ploss$, which answers how long the plasma would retain its stored energy if heating were switched off. This is benchmarked against the empirical IPB98(y,2) scaling law derived from a multi-machine experimental database

The simulation achieves H98y2 = 0.68, meaning the plasma retains heat at 68% of the level expected from the empirical H-mode database. The ITER Q = 10 design scenario requires H98y2 = 1.0, alongside fGW = 0.85 and $\beta n = 1.8$ simultaneously.Reason:

- **flat_factor = 1.0** over-flattens the core temperature at every sawtooth crash, keeping Te0 artificially low
- **TGLFNN underpredicts H-mode temperatures** — a known documented bias in the model itself
- **Current profile not settled** — only 150 s at 15 MA, but resistive diffusion takes 200–400 s
- **Density too low** — fGW = 0.74 vs design target 0.85, $\beta n = 1.27$ vs design target 1.8

4.4 Sawtooth Instabilities

The sawtooth instability is a periodic MHD crash that occurs when the central safety factor q_0 drops below 1.0 at the magnetic axis. When $q_0 < 1$, the **m=1, n=1 internal kink mode** becomes MHD unstable — field lines that complete exactly one poloidal turn per toroidal turn are resonant with this mode, and the resulting reconnection violently flattens the core temperature and current density profile. After the crash, q_0 rises back above 1, the profiles rebuild through ohmic and auxiliary heating, and the cycle repeats. This is distinct from the Kruskal-Shafranov condition, which governs edge kink stability and disruptions — not the internal sawtooth process.

In the simulation, act.ActorSawteethSource.flat_factor = 1.0 implements the **Kadomtsev complete reconnection model**: at every crash, the temperature and current profiles are fully flattened inside the entire q = 1 surface with no partial reconnection retained. This is the most aggressive possible sawtooth model. In reality, ITER sawteeth are expected to be incomplete reconnection events (flat_factor < 1.0), so this setting over-suppresses the central temperature at each crash, which is one of the four reasons H98y2 = 0.68 rather than the design target of 1.0. The sawteeth are visible in the **Vloop time trace** (red curve, right y-axis of the Ip/Vloop panel) as rapid small oscillations superimposed on the slowly varying inductive background — each spike corresponds to the sudden current redistribution during a crash event.

4.5 The Safety Factor Profile and q_{95}

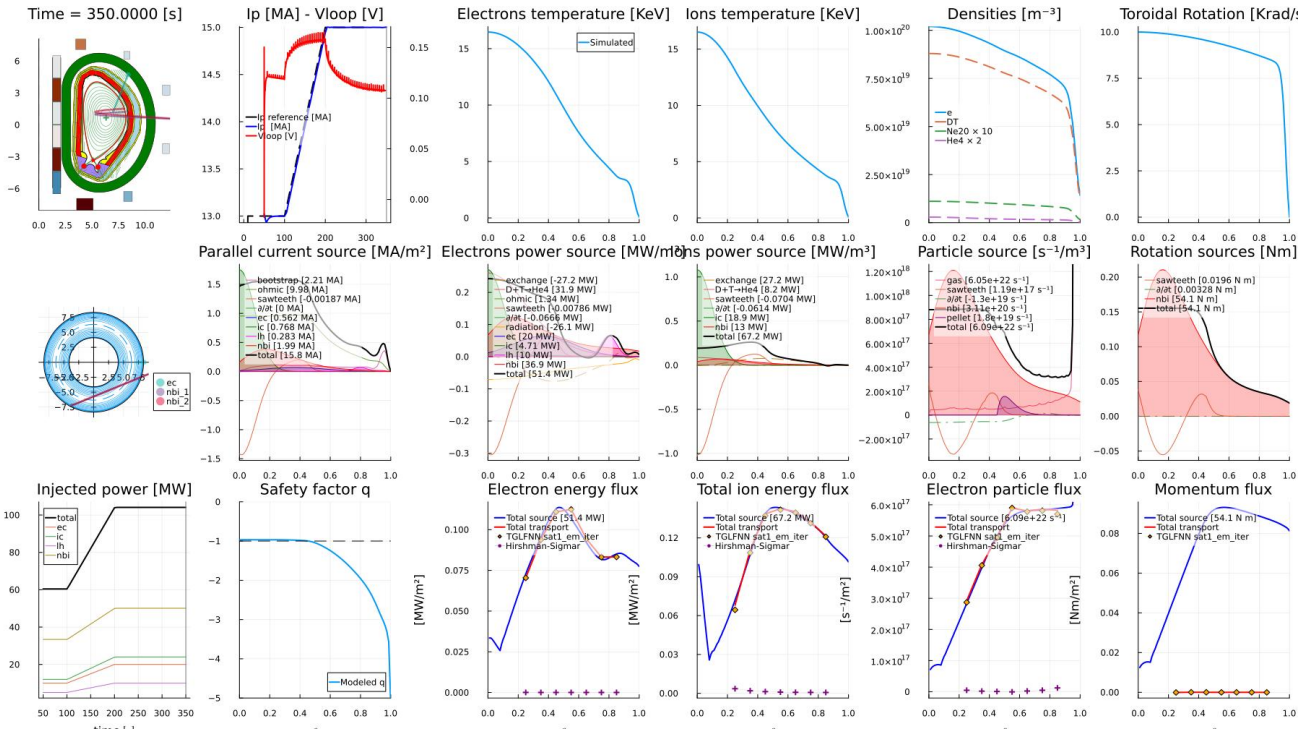
The safety factor $q(\rho) = rB\varphi/(RB\theta)$ measures the helical pitch of magnetic field lines. Its profile shape is crucial for MHD stability and transport. Key values:

- $q_0 < 1$ at the center → triggers sawtooth instabilities
- $q = 1$ surface → internal kink mode boundary
- $q = 2$ surface → neoclassical tearing modes (NTMs) can grow
- $q_{95} \approx -3.0$ → the value at 95% of the enclosed flux; ITER requires $|q_{95}| > 3$ for vertical stability

The negative sign of $q_{95} = -2.97$ is a sign convention the direction of toroidal current relative to toroidal field in ITER, physically $|q_{95}| = 2.97$, which is at the stability limit.

5. Plot-by-Plot Analysis

The plot_plasma_overview function produces a comprehensive 4×6 panel figure at each time step. The GIF animates this figure from t = 50 s to t = 350 s. Below is a detailed analysis of the final state (t = 350 s) and the evolution visible in the animation.



5.1 Panel 1: Plasma Cross-Section (Top-Left)

This shows the 2D poloidal cross-section of the tokamak at the current time. The nested closed curves are flux surfaces (surfaces of constant poloidal magnetic flux ψ). The outermost closed flux surface (LCFS) is the plasma boundary. The X-point at the bottom indicates the diverted configuration.

Evolution in GIF: The flux surface geometry evolves with Ip. As Ip increases from 13 to 15 MA, the poloidal field strengthens, the flux surfaces become more elongated, and the X-point configuration deepens. The colored dots show injection angles of EC, NBI_1, and NBI_2 beams.

The equilibrium shape (κ, δ) is largely fixed by the coil configuration (not actively controlled in this run with evolve_pf_active = false). The increasing Ip does shift the internal flux surface arrangement and the q-profile shape.

5.2 Panel 2: Ip [MA] - Vloop [V] (Time Trace)

This is the most important diagnostic panel. It shows:

- Blue dashed: Ip reference trajectory from dd.pulse_schedule — the target the controller is tracking
- Dark blue solid: Actual simulated Ip — how well the controller is following the reference
- Red solid: Loop voltage Vloop — the OH solenoid voltage that drives ohmic current

Evolution: Starting at t = 50 s with Ip = 13 MA, the current is flat until t = 100 s, then ramps to 15 MA by t = 200 s, then holds. The controller succeeds at t = 350 s, Ip = 15.0017 MA vs. reference 15.0 MA (0.01% error). The Vloop shows large oscillations (~0.05–0.15 V) corresponding to sawtooth crashes, superimposed on the slowly varying inductive component.

Vloop = η·Ip/A + L·dIp/dt + V_inductive. During ramp-up, the L·dIp/dt term dominates (inductive drive). During flat-top, the resistive term η·j dominates, requiring small but finite Vloop to maintain current against resistive dissipation. The sawtooth spikes in Vloop reflect rapid reconnection events.

5.3 Panel 3: Electron Temperature Profile Te(ρ) in KeV

The horizontal axis is the normalized toroidal flux coordinate ρ = √(Φ/Φ_LCFS), where ρ = 0 is the magnetic axis and ρ = 1 is the plasma boundary. The profile shows:

- Central Te_0 evolving from ~16 keV at t = 50 s to ~0 keV at t = 350 s
- Steep pedestal region near ρ = 0.85–1.0 (the H-mode pedestal)
- The 'Simulated' label indicates these are predictions from the transport model, not prescribed

Physics: The temperature profile shape is determined by the balance between heat flux (turbulent transport) and volumetric heat sources. The pedestal forms due to the suppression of turbulence in the strong ExB shear region at the plasma edge in H-mode.

5.4 Panel 4: Ion Temperature Profile Ti(ρ)

Ti_0 = 16.5 keV at t = 50 s, essentially equal to Te_0. The ion-electron equilibration rate Qei ∝ ne^2(Te-Ti)/(Te^(3/2)) is fast enough at ITER densities to keep Ti ≈ Te in the core.

5.5 Panel 5: Density Profiles n(ρ)

Four species are shown:

- ne (solid blue): electron density, peaks at 1.02×10^{20} m^{-3} and falls to ~5.9×10^{19} m^{-3} at the pedestal
- DT (dashed orange): deuterium-tritium fuel ions, dominant species
- Ne20 × 10 (green, scaled up): neon impurity used for seeding/radiation control
- He4 × 2 (purple, scaled): helium ash from fusion reactions, must be exhausted by the divertor

5.6 Panel 6: Toroidal Rotation $\omega_0(\rho)$

The toroidal rotation is held fixed at ω_0 = 10 krad/s throughout the simulation (not evolved). This is a simplification - in reality, NBI torque drives toroidal rotation, competing with neoclassical toroidal viscosity (NTV) and magnetic braking. The rotation profile falls from 10 krad/s at the center to near zero at the edge, reflecting the assumed boundary condition of zero rotation at ρ = 1.

Reason: Rotation affects turbulent transport through ExB shear stabilization — the E×B shearing rate γExB = (ρ/q)(∂ωE/∂ρ) suppresses turbulence when γExB > γ_max_linear. At ITER scale, ExB shear is modest and not the dominant confinement mechanism.

5.7 Panel 7: Parallel Current Source j_ // (ρ)

This panel decomposes the total parallel current density (black, total ~13.6 MA at t = 50 s → 15.8 MA at t = 350 s) into its constituent sources:

- Bootstrap [2.07→2.21 MA]: Self-generated neoclassical current, peaks off-axis where pressure gradient is steepest
- Ohmic [9.65→9.39 MA]: OH solenoid-driven current, broad central deposition
- EC [0.234 MA]: Electron Cyclotron current drive, localized at ρ ≈ 0.37
- IC [0.342 MA]: Ion Cyclotron current drive via mode conversion
- LH [0.133 MA]: Lower Hybrid current drive, typically off-axis

- NBI [1.16→1.19 MA]: Neutral beam current drive from fast ion momentum
- Sawteeth [~0]: Sawtooth-induced current redistribution term

Panels 8 & 9: Power Source Profiles

Electron Power Source (Panel 8) and Ion Power Source (Panel 9) show the volumetric heating deposition profiles [MW/m³] for each heating channel:

- Exchange [-ve MW electrons, +ve MW ions]: Electron-ion energy transfer. Negative for electrons because they lose energy to the colder ions in the outer region; positive for ions in the core where Te > Ti
- D+T→He4 : Alpha particle heating of electrons, centrally peaked
- Ohmic [1.34 MW]: Resistive heating $\eta \cdot j^2$, small in hot plasma
- Sawteeth
- EC [10 MW]
- IC : Ion cyclotron minority heating, peaked in core
- NBI: Broad deposition from beam slowing-down, primarily to ions
- Radiation [-26.1 MW electrons]

Panel 10: Particle Source Γ(ρ) [s^{-1}/m^3]

Shows the volumetric particle sources:

- Gas puffing [6.05×10^{22} s^{-1}]: Dominant fueling source at the edge
- Pellet injection [1.8×10^{18} s^{-1}]: Deep fueling at higher ρ
- Sawteeth [1.8×1017 s^{-1}]: Negligible particle redistribution during crashes
- NBI [3.11×10^{20} s^{-1}]: Beam ionization deposits particles in the core

The total particle source must balance the particle flux to the wall (diverted to the divertor) in steady state. The current simulation uses TGLFNN and Hirshman-Sigmar models for particle transport.

5.11 Panel 11: Rotation Sources [Nm/m³]

Shows torque density from each mechanism. NBI dominates [54.1 N·m total], providing co-injected torque that drives the plasma to rotate toroidally. The sawtooth term redistributes angular momentum during crashes. Rotation affects turbulent transport and MHD stability.

5.12 Panel 12: Injected Power Time Evolution

The bottom-left panel shows the time evolution of total injected power [MW] from t = 50 s to t = 350 s:

- NBI (yellow)
- EC (green)
- IC (dark green)
- LH (purple)
- Total (black)l

This directly reflects the ini waveforms: (1 + ramp((t-100)/100)) × P_0, which gives a linear ramp from P_0 to $2P_0$ over t = 100–200 s. The flat segments at t = 50–100 s and t = 200–350 s correspond to the flat portions of the ramp function.

5.13 Panel 13: Safety Factor q(ρ)

The q profile shows:

- Central $q_0 < 1$ (below the q=1 dashed line) → sawtooth active region
- Monotonically increasing outward (no reversed shear in this scenario)
- $q_{95} = -2.97$ (sign convention; $|q_{95}| \approx 3.0$ at the plasma edge)

The q profile evolves slowly as the current diffuses inward. At t = 50 s (simulation start), the current profile is not yet fully relaxed from the ramp-up. By t = 350 s, the ohmic component has partially diffused inward, slightly reducing q_0 and broadening the sawtooth-affected region. The q = 2 surface is at approximately ρ ≈ 0.75–0.80, which could be susceptible to NTM seeding - but NTM physics is not included in this FUSE run.

5.14 & 5.15: Electron and Ion Energy Flux [MW/m²]

These flux-matching panels are crucial for understanding confinement quality. They show three curves:

- Blue solid (Total source): The cumulative power deposited inside radius ρ — what the transport must carry outward
- Red dashed (Total transport): The predicted turbulent flux from the TGLFNN neural-network transport model
- Diamond markers (TGLFNN sat1_em_iter): Stiff transport model evaluation points
- Circle markers (Hirshman-Sigmar): Neoclassical benchmark

Convergence is achieved when Total source ≈ Total transport at each ρ. The fact that the curves match well validates that the transport solver has converged. at t = 350 s more energy goes to ions because NBI at 1 MeV primarily heats ions.

5.16: Electron Particle Flux [s^{-1}/m^2]

Shows the radial particle transport and matches the total particle source with the outward diffusive flux. The particle confinement is coupled to energy confinement through the thermal diffusivities. Poor particle confinement would lead to density peaking loss.

5.17: Momentum Flux [N·m/m²]

The momentum flux is driven by NBI torque (54.1 N·m total) and must balance with neoclassical momentum transport. This panel validates the rotation model. Weak momentum transport in the ITER core means NBI can sustain modest rotation (10 krad/s) even at ITER scale.